

Software engineer with extensive experience architecting and deploying scalable backend systems, recommendation platforms, and data-intensive applications across enterprise environments. I collaborate closely with diverse teams while actively adapting to agentic AI-driven development, using it to accelerate coding, explore solutions, and improve developer efficiency. Passionate about building efficient, measurable systems that translate technical solutions into business impact.

SKILLS

Languages	Python, Java, SQL, Scala.
ML Frameworks	Sci-kit-Learn, Keras, Tensor Flow.
Developer Tools	Git, Docker, AWS, Google Cloud Platform, $\LaTeX$.
Libraries	pandas, NumPy, Matplotlib.

TECHNICAL EXPERIENCE

Career Break / Relocation — Canada **Oct 2024 – Present**
• Relocated to Canada after my last role and used the transition period to focus on job search, skill development, and personal commitments.

Software Engineer **Oct 2022 – Oct 2024**
Monetate Inc *Austin, TX, USA*
• Owned the design and deployment of multiple recommendation systems (collaborative and content-based) that support personalized product discovery across enterprise e-commerce clients.
• **Dynamic Slotting**
– Architected and implemented a dynamic slotting algorithm that intelligently aggregated and ranked outputs from multiple recommendation models in real time.
– Replaced static placement logic with context-aware weighted scoring based on user behavior and product affinity signals.
– Increased click-through and conversion rates by 10%, driving measurable revenue uplift across multiple enterprise clients.
• **Cross-Placement Relevance & De-duplication**
– Engineered a high-performance cross-placement de-duplication layer to eliminate redundant product recommendations across personalized modules.
– Introduced diversity-aware ranking constraints while preserving relevance and maintaining low-latency serving.
– Improved recommendation quality, enhanced perceived personalization and increased customer clickrate by 20%.
• **Miscellaneous**
– Led client-facing beta rollouts for new algorithms, coordinating testing, debugging, and staged releases across multiple customer environments.
– Designed and analyzed A/B/n experiments on live production traffic, using KPIs such as Conversion Rate, Revenue per Session, and Bounce Rate to drive data-backed product decisions.
– Built internal and client-facing analytics dashboards (Looker, Streamlit, Matplotlib), which improved visibility into experiment results and accelerated stakeholder decision-making.
– Reduced data pipeline costs and query latency by identifying and optimizing inefficient Snowflake SQL queries, improving system reliability and lowering operational overhead.

Software Engineer **Nov 2021 – Oct 2022**
KIBO Software Inc *Austin, TX, USA*
• **“Similar Items” Recommendation System**
– Architected and deployed a scalable “Similar Items” recommendation service leveraging product metadata (color, category, price, attributes) to power context-aware cross-sell experiences across enterprise e-commerce storefronts.
– Designed similarity computation and ranking pipelines with configurable weighting and diversity constraints to balance relevance, catalog coverage, and business objectives.
– Improved cross-sell engagement and revenue contribution by 20% across enterprise clients.
• **Omni-channel Data Integration for Personalization**
– Led end-to-end integration of offline purchase data into real-time recommendation pipelines, designing data ingestion, transformation, and feature alignment workflows to unify online and in-store behavioral signals.
– Enabled richer user representations that improved personalization accuracy, driving a 25–30% increase in client conversion metrics.
• **Miscellaneous**
– Partnered closely with product managers and client teams to translate business requirements into production-ready features.
– Resolved high-priority bugs and performance bottlenecks, reducing client-reported incidents and improving overall system stability.
– Contributed to the migration of the codebase from Python 2 to Python 3, resolving compatibility issues and modernizing legacy modules to ensure long-term maintainability.

Senior Systems Engineer

Infosys Ltd

Nov 2015 – Jun 2018

Chennai, India

- Designed and developed a full-stack internal workflow automation platform to streamline ETL processes, reducing manual intervention and improving data availability for analytics teams.
- Built backend data processing modules in Python and SQL to orchestrate scheduled transformations, validation checks, and error handling for large client datasets.
- Completed comprehensive training in Python, SQL and Java and contributed to rapid feature delivery on client projects.

EDUCATION

Master of Science in Computer Science

University of Georgia

Aug 2018 - Aug 2021

Athens, GA, USA

- **Courses** : Computer Vision, Data Science, Distributed Computing Systems, Database Management Systems, Algorithms, Cloud Computing.
- **Masters Thesis** : Object Detection in 3D Coral Ecosystem Maps from Multiple Image Sequences
- **Graduate Research Assistant** : Center for Applied Isotope Studies, UGA.
 - Developed a Python-based desktop ETL and visualization tool for isotope analysis, streamlining data interpretation for research teams.
 - Prototyped predictive modeling modules, laying the groundwork for advanced analytics integration in scientific workflows.
- **Graduate Teaching Assistant** : CSCI-4300 Web Programming and CSCI-1100L Topics in Computing Lab.
 - Mentored 100+ students in Web Programming and Topics in Computing, supporting mastery of Python, HTML/CSS, JavaScript and SQL.
 - Developed lab assignments, automated grading tools, and supplemental teaching materials to streamline instruction and improve student outcomes.

Bachelor of Engineering in Computer Science

Anna University

Aug 2011 - May 2015

India

- **Courses** : Theory of Computer Science, Data Structures, Algorithms, Object Oriented Programming, Computer networks, Discrete Maths.

RESEARCH AND PUBLICATIONS

Object Detection in 3D Coral Ecosystem Maps from Multiple Image Sequences

University of Georgia

- Proposed and implemented a novel algorithm to detect and localize individual coral entities in 3D ecosystem maps — converting 2D image detections into accurate 3D objects for ecological mapping.
- Combined 2D object detectors (e.g., Faster R-CNN, RetinaNet) with camera-calibration parameters and ray-tracing to project 2D detections into a 3D map generated via SLAM, enabling volumetric reef analysis.
- Benchmarked results against established 3D methods (Frustum ConvNet, PointNet variants), highlighting trade-offs and performance characteristics.
- Published at the 26th International Conference on Pattern Recognition (ICPR 2022).
- paper : [Object Detection in 3D Coral Ecosystem Maps from Multiple Image Sequences](#)